Navya Popuri

ch24btech11022@iith.ac.in | navyarsh.popuri@gmail.com | +91 9380996681 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.42
JINDAL VIDYA MANDIR, VIDYA NAGAR	Class XII (CBSE)	2024	96.60%
JINDAL VIDYA MANDIR, VIDYA NAGAR	Class X (CBSE)	2022	97.40%

WORK EXPERIENCE

Product Strategy Intern | GTM Unbound

- Conducted ICP research and founder profiling for **100+ startups**, segmenting by product category (AI/SaaS/B2B), GTM maturity, and India-US expansion readiness to align founders with investor and event goals

- Automated the **internship's ICP research** and founder-screening workflow by building **FounderLens**, an **agentic AI** tool , generating **Strong/Partial/Not-a-Fit** recommendations against investor theses.
- **Reduced manual research time** by implementing an **anti-hallucination** multi-query retrieval pipeline.

RELEVANT COURSES AND CERTIFICATIONS

CS and AI/ML Courses: Introduction to Programming, Introduction to Artificial Intelligence

Mathematics Courses: Calculus-1&2, Differential Equations, Elementary Linear Algebra, Applied Maths for Chemical Engineering, Numerical Methods

Certificates: IBM Data Science Professional Certificate-IBM (Coursera), Machine Learning Specialization-Stanford University & DeepLearning.AI (Coursera), McKinsey Forward Program (McKinsey & Company)

SKILLS

Web & Tools : React, Flask, Jupyter Labs, Streamlit, Excel, Figma, Canva, PowerPoint

Programming & ML: Python, C,C++, SQL, Data Structures and Algorithm, NumPy, Pandas, Scikit-learn, XGBoost, CatBoost, SHAP, PowerBI

PROJECTS

Parameter Estimation for Li-Rinzel Calcium Oscillation Model

- Built an end-to-end Python pipeline to estimate IP3 and k3 parameters of the nonlinear Li-Rinzel ODE model against noisy experimental Ca2+ time-series data, including baseline correction and normalization.
- Designed a **900-run** Monte Carlo search (30×30 grid) combining KL divergence and oscillation-frequency mismatch into a composite fitness score, achieving RMSE of **0.31** and **KL = 0.21**
- Generated 2D heatmaps and 3D error-surface visualizations to characterize parameter sensitivity, revealing a smooth, convex search landscape

Prime Explorer

- Implemented advanced algorithms in number theory and numerical analysis (Miller-Rabin, Lucas-Lehmer)
- Optimized performance for extreme-scale inputs (up to **30000-digit integers**) using bitwise operations
- Designed a modular full-stack system with a React frontend and Flask backend featuring inputs and error handling.

LRU Cache Engine

- Implemented an **O(1)** LRU cache in C++ using a hash map + doubly linked list with sentinel nodes, guaranteeing constant-time get/put while eliminating head/tail edge cases
- Wrote **22 unit tests** in a custom zero-dependency test harness, covering eviction order, duplicate-key updates, invalid-capacity handling, and a 1**0K-operation stress test** to verify correctness under sustained load.

Autocomplete & Spell-Checker

- Implemented a Trie for O(k) prefix search over a **20,000**-word dictionary, ranking completions by frequency then alphabetically, with **enter-to-search** and **live-keystroke** modes, plus a DP-based **edit-distance** algorithm.
- Built a Levenshtein-distance spell-checker returning the optimal **top-5** dictionary matches (within edit distance 2) when no autocomplete hits are found, mirroring real-world autocomplete/spell-check engines.

Olist E-Commerce Product Analytics

- Built an end-to-end **SQL analytics pipeline** using **CTEs** and **window functions** on **100K+ real orders** to analyze conversion funnels, **customer retention cohorts**, and **RFM-based customer segments**.
- Performed **hypothesis testing** using **chi-square and Welch's t-tests** to evaluate the impact of delivery delays, finding no significant effect (**p=0.3**) on repeat purchases but a decline in customer satisfaction (**4.29 to 2.57**)

Customer Churn Prediction Model

- Identified churn as a high-cost problem (5-25× costlier to acquire vs retain); optimized recall from **0.50 to 0.79.**
- Delivered personalized, SHAP-driven retention recommendations through an interactive Streamlit dashboard
- Built a **fault tolerant flow**, GenAI-generated by default, rule based fallback on API failure for reliability.

POSITIONS OF RESPONSIBILITY

Head of Ideation | Entrepreneurship Cell (E-Cell), IIT Hyderabad

- Leading a team of **11** to drive ideation initiatives and startup engagement across IITH community.

Consulting Coordinator | Finance and Consulting Club (FCC), IIT Hyderabad

- Selected from **150+** applicants to lead initiatives within a 500+ member consulting community

ACHIEVEMENTS & EXTRACURRICULAR

- **Amex Strategy Track Campus Case Challenge:** Ranked in the **top 11%** among **17,000+ teams**; built a scoring model achieving **88% accuracy** to measure customer profitability on credit card data.

- **First Dan (Black Belt), Shotokan Karate** World Traditional Shotokan Karate Federation (WTSKF), 2018